Fiscal Year:	FY 2009	**Task Last Updated:**	FY 05/18/2009
PI Name:	Fuller, Charles A. Ph.D.		
Project Title:	Artificial Gravity and Light as Countermeasures for Circadian Dysfunction in Altered Gravity Environments		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	cafuller@ucdavis.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	530-752-2979
Organization Name:	University of California, Davis		
PI Address 1:	Section of Neurobiology, Physiology & Behavior		
PI Address 2:	One Shields Avenue, 196 Briggs Hall		
PI Web Page:			
City:	Davis	**State:**	CA
Zip Code:	95616-5270	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	05/15/2004	**End Date:**	06/01/2011
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JSC
Contact Monitor:	Shea, Cami1e	**Contact Phone:**	281-244-2017
Contact Email:	shea@dsls.usra.edu		
Flight Program:			
Flight Assignment:	NOTE: Received NCE until 6/01/2011, per C. Guidry/JSC (9/2010) NOTE: Received NCE until 5/31/2010, per J. Dardano/JSC (4/2009) NOTE: Funding reinstated to 06/01/2009, per J. Cohen (HQ, 8/2006)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Robinson, Edward Ph.D. (University of California, Davis)		
Grant/Contract No.:	NNJ04HF44G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The Circadian Timing System (CTS) is an important temporal organizer controlling both the physiology and behavior of organisms. The CTS has been shown to be sensitive to altered force environments (AFE) in both the microgravity of space flight and in hypergravity produced by centrifugation. Evidence from several species shows that AFE produces changes in the period of the circadian clock (tau). Evidence from rodents also shows that AFE can phase-shift and synchronize the circadian pacemaker. Further, sensitivity to light exposure, the most potent synchronizer of the CTS, appears to be impaired in AFE, as shown by both molecular and physiological evidence. Thus, circadian regulation in AFE may result from interactions of G and light inputs to the CTS. It is important to understand how AFE alters the CTS response to light exposure to ensure that countermeasures for CTS dysfunction such as timed bright light are equally effective in AFE. We will use AFE produced by centrifugation to determine if the phase-shifting responses to light are impaired in AFE. Countermeasures for circadian dysfunction may differ from those suitable at 1G. We will test a light countermeasure for circadian dysfunction, with defined spectral content, duration and schedule, to determine how its efficacy is affected by AFE. Timed AFE exposure by centrifugation is a proposed countermeasure for changes in cardiovascular, skeletal, muscle and other systems in space flight. Pulses of AFE, however, can entrain and phase shift the CTS in rats, which possess similarities in CTS inputs to rhesus and humans. Of concern is that the use of a timed AFE countermeasure could potentially compromise CTS function, or even conflict with circadian countermeasures, by inappropriately phase-shifting the CTS. It is important to understand how acute AFE exposure at different times of day alters circadian phase and period in a diurnal primate. Impaired phase shifting by light would impact adjustment to new schedules as well as entrainment to either 24-h or non-24-h days. Conversely, if AFE is capable of phase shifting or entraining the primate CTS, AFE could also serve as a countermeasure for circadian dysfunction.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This program will characterize disturbances of sleep in rhesus monkeys. The rhesus is a well established model for human physiology in many disciplines and is showing promise as a model of human sleep disturbances. This study will directly relate to issues of sleep and circadian rhythm disturbance as a result of occupational factors such as shift work or other irregular schedules and jet lag. By establishing countermeasures which improve sleep and performance in both hypergravity and normal gravity, we will be advancing what is known about potential treatment of sleep problems on earth. Further, optimization of countermeasures for space flight will likely involve controlled light exposure to help ensure adequate sleep. This will need to be based on an understanding of how the primate visual system responds to light, in particular to spectral content and intensity. Recent findings of that non-visual photoreception is linked to the CTS, and likely to sleep consolidation as well, underscores the importance of quantitative development of specific light countermeasures. Understanding sleep disturbances and the responses of the CTS, including sleep and performance to altered gravity is also important to extending the ability of humans to function effectively in stressful terrestrial environments.
Task Progress:	This research program has made significant progress in examination of the mechanisms underlying the response of the circadian timing system (CTS) of primates to altered gravity. We have initiated new studies on the interaction of altered gravity and light sensitivity on the CTS. We have examined changes in circadian rhythms, sleep and performance to controlled lighting environments, with and without the perturbation of hypergravity produced by centrifugation. Several improvements have been made to the telemetry system and to the centrifuge housing and data collection system. In addition we have collaborated on a revision of the Psychomotor Test System for performance metrics, which also serves as enrichment by providing an operant task to the animals. In studies of blue light as a potential circadian countermeasure we have seen that the effects of blue light on sleep and alerting depend on the time of day of presentation. We used non-24-h light dark cycles of alternating 3-h light and 3-h dark periods (LD and DL 3:3) to present light and dark at all times of the 24-h day. Increased sleep was seen following blue light presentation during the day and a phase-dependent alerting effect was seen following blue light exposure early in the subjective night. Blue light was tested as a substitute for white light in a conventional 24-h day of 12-h light and 12-h dark (LD 12:12) to provide a worst-case test of augmented blue wavelengths, known to have the maximum effect on circadian timing. In this study, total sleep time and individual sleep stages did not differ significantly in blue light with the exception of increased rapid eye movement (REM) sleep. However, phase advances in sleep and brain temperature rhythms were seen in blue light, indicating phase re-alignment (advance) during the 24-h day. We have completed a renovation of our centrifuge facility for suitability with larger rhesus and can now evaluate how these changes interact with those induced by an altered gravitational field. Data analysis and manuscript preparation for the above studies is ongoing.
Bibliography Type:	Description: (Last Updated: 12/08/2025)
Abstracts for Journals and Proceedings	Robinson EL, Fuller CA. "Blue Light Phase Advances Entrained Circadian Rhythms of Rhesus Monkeys." Presented at the NASA HRP Investigators' Workshop, Houston, February 2008. HRP Investigators' Workshop, Houston, February 2008. , Feb-2008
Abstracts for Journals and Proceedings	Robinson EL, Fuller, CA. "Blue Light Phase Advances Entrained Rhythms of Brain Temperature and Sleep in Rhesus Monkeys." Presented at Experimental Biology, San Diego, CA, April 2008. Experimental Biology, Abstract 1222.4, April 2008. , Apr-2008
Abstracts for Journals and Proceedings	Robinson EL, Fuller PM, Fuller CA. "Blue Light Affects Timing of Sleep in Rhesus Monkeys but not Sleep Homeostasis." European Sleep Research Society, Edinburg, Scotland, Sept. 2008. European Sleep Research Society, Sept. 2008. , Sep-2008
Articles in Peer-reviewed Journals	Hsieh K-C, Robinson EL, Fuller CA. "Sleep architecture in unrestrained rhesus monkeys (Macaca mulatta) synchronized to 24-hour light-dark cycles." Sleep. 2008 Sep 1;31(9):1239-50. PMID: 18788649 , Sep-2008